NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 2.

NAMES: _____ DATE OF BIRTH: _____

SCHOOL/INSTITUTION: _____ BRANCH: _____

CLASS: _____ GENDER: _____ DISTRICT: _____

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

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PLEASE, NOTE CAREFULLY:

1. **WRITE YOUR ANSWERS IN THE SPACES PROVIDED. MORE SPACE IS AVAILABLE AT THE END OF THIS PAPER.**
2. Date: Saturday, 29th June, 2019.
3. Time: Exactly 2 hours and 15 Minutes (from 9 : 00am to 11 : 15am)
4. Type:
   (a) This paper consists of two sections and each section carries 50 marks.
   (b) Attempt as many questions as you can. The total marks scored from all the questions you attempt will be your final score.
   (c) No marks will be awarded for an answer if NO clear and logical layout of the working is shown.
5. **Indicate your names, date of birth, gender, class, school and district on all your answer sheets**.
6. All diagrams/figures in this contest are not to scale.
7. Immediately the paper is done, the answer script(s) must be forwarded to:

   David Ddumba Walakira,
   Contest Coordinator, Uganda Mathematical Society,
   c/o Department of Mathematics, Makerere University,
   P.O. Box 7062, Kampala.
   Telephone contact: $+256-414-540692$.

   Answer scripts should be received **latest, Monday 1st July,** 2019.
8. **ALL** *participants who qualified and sat Paper* 2 *are cordially invited to the certificate and/or prize giving ceremony on Saturday* 27*th July,* 2019 ***at 8:00am*** *at Makerere University, Kampala.*
9. National Mathematics Contest 2020 Paper 1 will be done on Saturday 4th April, 2020 at various centres.
10. **UGANDA MATHEMATICAL SOCIETY** wishes you success in this year's contest.

SECTION A : 5 marks each

Qn. 1. Which of these is the smallest? 1.0101, 1.0011, 1.0110, 1.1001, 1.1100

Qn. 2. $\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}$ equals (choose right answer).

A. $3\frac{1}{3}$ B. $8+\frac{1}{3}$ C. $\frac{3}{8}$ D. $8+3$ E. $\frac{1}{3}\times 8$

Qn. 3. Place each of the four numbers $3, 4, 6, 7$ in the box so that when the fractions are added, the result is the largest. Find this largest sum.

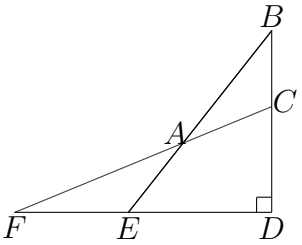
$\frac{\square}{\square}+\frac{\square}{\square}=$

Qn. 4. Ken rode the first 12km of his 36km bicyle trip before lunch. What fraction of his overall trip remains to be completed after lunch?

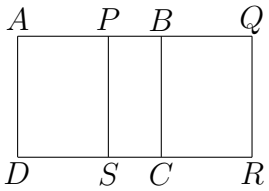
Qn. 5. Five children had dinner. Charles ate more than Maria. Benon ate less than Ken. Ken ate less than Maria but more than Tina. Which child ate the second most?

Qn. 6. The capacity of a swimming pool is 4000 litres. Stella starts filling the empty swimming pool with water at a rate of 20 litres per minute. The swimming pool develops a leak after 20 minutes and loses water at 2 litres per minute. Beginning from the time when Stella starts filling the empty swimming pool, how long does it take to fill the swimming pool?

Qn. 7. In the figure below, angle $BED = 60°$, and angle $FCD = 48°$. What is the size of angle CAE?



Qn. 8. Two identical squares $ABCD$ and $PQRS$ have side length 12cm. They overlap to form a 12cm by 20cm rectangle $AQRD$. What is the area of rectangle $PBCS$?



Qn. 9. Roy buys a pizza cut into 12 equal parts. He then cuts each part into 2 equal parts. If he eats 3 of these pieces, what fraction of the pizza does he eat?

Qn. 10. For every 3 chocolates that Claire buys at the regular price, she buys a fourth chocolate at Shs. 250. Claire buys 12 chocolates for Shs. 6150. What is the regular cost of one chocolate?

SECTION B : 10 **marks each**

Qn. 11. How many different (x, y) combinations can be formed using numbers from the list

$$\{1,\ 2,\ 3,\ 4,\ 5, \ldots,\ 19,\ 20\}$$

such that $x < y$ and $x + y$ is even?

Qn. 12. Rose assigns every letter of the alphabet a different whole number value. She finds the value of a word by multiplying the values of its letters together. For example, if D has a value of 10 and I has a value of 8, then the word DID has value of $10 \times 8 \times 10 = 800$. The table shows the value of some words. What is the value of the word MATH?

Word	Value
TOTE	18
TEAM	168
MOM	49
HOME	70
MATH	?

Qn. 13. In the magic square, all rows, columns and diagonals have the same sum. The magic square uses each of the numbers from -6 to $+2$ only. What is the value of $a + b + c + d + e$?

$+1$	a	d
-4	b	c
-3	e	-5

Qn. 14. Three pineapples are weighted two at a time in all possible ways. The weights of the pairs of pineapples are 12 kg, 13 kg and 15 kg. What is the weight of the least heavy pineapple?

Qn. 15. Chris and Farida want to buy the same book. Chris has $\frac{3}{4}$ of the cost of the book and Farida has half of the cost of the book. If the book was Shs. $30,000$ cheaper, then together they would have exactly enough money to buy 2 copies of the book. What is the original price of the book?

NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 2 SOLUTIONS.

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SECTION A : 5 marks each

Qn. 1. Which of these is the smallest? 1.0101, 1.0011, 1.0110, 1.1001, 1.1100

Solution :

Using tenths digits, we eliminate 1.1001 *and* 1.1100

Using hundreds digits, we eliminate 1.0101 *and* 1.0110

The smallest is 1.0011. ■

Qn. 2. $\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}$ equals (choose right answer).

A. $3\frac{1}{3}$ B. $8+\frac{1}{3}$ C. $\frac{3}{8}$ D. $8+3$ E. $\frac{1}{3}\times 8$

Solution : *Add* $\frac{1}{3}$ *eight times, so E* ■

Qn. 3. Place each of the four numbers $3, 4, 6, 7$ in the box so that when the fractions are added, the result is the largest. Find this largest sum.

$\frac{\square}{\square}+\frac{\square}{\square}=$

Solution : *The largest sum is attained when the numerators are the largest and denominators smallest. Combine largest numerator (*7*) and smallest denominator (*3*), and* 6 *with a* 4 *such that*

$$\frac{7}{3}+\frac{6}{4}=\frac{(4\times 7)+(3\times 6)}{12}=\frac{28+18}{12}=\frac{46}{12}=\frac{23}{6}$$

■

Qn. 4. Ken rode the first 12km of his 36km bicyle trip before lunch. What fraction of his overall trip remains to be completed after lunch?

Solution : *The remaining trip is* $36-12=24km$. *The fraction is* $\frac{24}{36}=\frac{2}{3}$ ■

Qn. 5. Five children had dinner. Charles ate more than Maria. Benon ate less than Ken. Ken ate less than Maria but more than Tina. Which child ate the second most?

Solution : *Let B (Benon), C (Charles), K (Ken), M (Maria), T (Tina)*

$$B < K < M < C$$
$$T < K$$

Maria ate the second most. ■

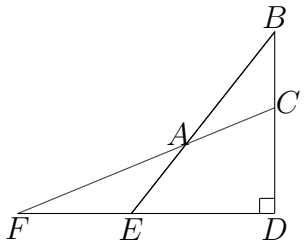
Qn. 6. The capacity of a swimming pool is 4000 litres. Stella starts filling the empty swimming pool with water at a rate of 20 litres per minute. The swimming pool develops a leak after 20 minutes and loses water at 2 litres per minute. Beginning from the time when Stella starts filling the empty swimming pool, how long does it take to fill the swimming pool?

Solution : *By the 20th minute, the pool has* $20 \times 20 = 400$ *litres, requiring* $4000 - 400 = 3600$ *litres to be filled.*

Due to leakage of 2 *litres per minute, the filling rate is now* $20 - 2 = 18$ *litres per minute. Thus to fill the pool (remaining* 3600 *litres) will take* $\frac{3600}{18} = 200$ *minutes.*

The total time is then $20 + 200 = 2200$ *minutes, or 3hrs:40min* ■

Qn. 7. In the figure below, angle $BED = 60°$, and angle $FCD = 48°$. What is the size of angle CAE?

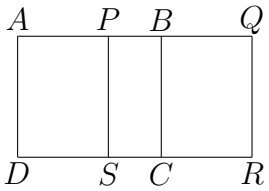


Solution : *Let* $\angle CAE = x°$. *Thus*

$$\begin{aligned} x + 60° + 48° + 90° &= 360° \\ x + 198° &= 360° \\ x &= 360° - 198° \\ x &= 162° \end{aligned}$$

■

Qn. 8. Two identical squares $ABCD$ and $PQRS$ have side length 12cm. They overlap to form a 12cm by 20cm rectangle $AQRD$. What is the area of rectangle $PBCS$?



Solution : *The area of a rectangle* $PBCS = \overline{PS} \times \overline{PB}$

Given

$$\overline{AQ} = 20cm; \quad \overline{AB} = \overline{PQ} = 12cm; \quad \overline{PS} = 12cm; \quad \overline{PB} = \overline{PQ} - \overline{BQ}$$

But

$$\overline{BQ} = \overline{AQ} - \overline{AB} = 20 - 12 = 8cm$$

Therefore, $\overline{PB} = 12 - 8 = 4$ *cm. Thus, the area of the rectangle is* $12 \times 4 = 48cm^2$.
■

Qn. 9. Roy buys a pizza cut into 12 equal parts. He then cuts each part into 2 equal parts. If he eats 3 of these pieces, what fraction of the pizza does he eat?

Solution : *Total number of pieces after cutting each of the* 12 *pieces into* 2 *equal parts is* $12 \times 2 = 24$ *pieces. Roy ate* 3 *of the* 24 *pieces; Thus fraction eaten is* $\frac{3}{24} = \frac{1}{8}$ ■

Qn. 10. For every 3 chocolates that Claire buys at the regular price, she buys a fourth chocolate at Shs. 250. Claire buys 12 chocolates for Shs. 6150. What is the regular cost of one chocolate?

Solution : *For a total of* 12 *chocolates, Claire bought three* 4th *chocolates, each at Shs.* 250. *Thus for these three, she spent* $250 \times 3 = 750$ *shillings.*

The cost of the other $12 - 3 = 9$ *chocolates was* $6,150 - 750 = Shs.5,400$. *Therefore, each of the other* 9 *chocolates cost Shs.*$\frac{5,400}{9} = Shs.600$ ■

SECTION B : 10 marks each

Qn. 11. How many different (x, y) combinations can be formed using numbers from the list

$$\{1,\ 2,\ 3,\ 4,\ 5, \ldots,\ 19,\ 20\}$$

such that $x < y$ and $x + y$ is even?

Solution : *To get $x + y$ even; either both x and y are even or both are odd.*

1. *If both are even:*
   - *If $x = 2$, then y can be $\{4, 6, 8, 10, 12, 14, 16, 18, 20\}$ giving 9 different (x, y) combinations such that $x < y$ and $x + y$ is even.*
   - *If $x = 4$, then y can be $\{6, 8, 10, 12, 14, 16, 18, 20\}$ giving 8 different (x, y) combinations such that $x < y$ and $x + y$ is even.*
   - *If $x = 6$, then y can be $\{8, 10, 12, 14, 16, 18, 20\}$ giving 7 different (x, y) combinations such that $x < y$ and $x + y$ is even.*
   - *and so on like that until*
   - *If $x = 18$, then y can be $\{20\}$ giving 1 different $(x, y) = (18, 20)$ combinations such that $x < y$ and $x + y$ is even.*

   Between 1 and 20, there are 9 even numbers less than 20, that is $\{2, 4, 6, 8, 10, 12, 14, 16, 18\}$

Even Nos	*2*	*4*	*6*	*8*	*10*	*12*	*14*	*16*	*18*	*Total Combinations*
No. of Combinations	*9*	*8*	*7*	*6*	*5*	*4*	*3*	*2*	*1*	*45*

2. *If both are odd. x must be odd and y odd.*
   - *If $x = 1$, then y can be $\{3, 5, 7, 9, 11, 13, 15, 17, 19\}$ giving 9 different (x, y) combinations such that $x < y$ and $x + y$ is even.*
   - *If $x = 3$, then y can be $\{5, 7, 9, 11, 13, 15, 17, 19\}$ giving 8 different (x, y) combinations such that $x < y$ and $x + y$ is even.*
   - *If $x = 5$, then y can be $\{7, 9, 11, 13, 15, 17, 19\}$ giving 7 different (x, y) combinations such that $x < y$ and $x + y$ is even.*
   - *and so on like that until*
   - *If $x = 17$, then y can be $\{19\}$ giving 1 combination such that $x < y$ and $x + y$ is even.*

   Between 1 and 20, there are 9 odd numbers less than 19, that is $\{1, 3, 5, 7, 9, 11, 13, 15, 17\}$

Odd Nos	*1*	*3*	*5*	*7*	*9*	*11*	*13*	*15*	*17*	*Total Combinations*
No. of Combinations	*9*	*8*	*7*	*6*	*5*	*4*	*3*	*2*	*1*	*45*

The total number of combinations is $45 + 45 = 90$. ■

Qn. 12. Rose assigns every letter of the alphabet a different whole number value. She finds the value of a word by multiplying the values of its letters together. For example, if D has a value of 10 and I has a value of 8, then the word DID has value of $10 \times 8 \times 10 = 800$. The table shows the value of some words. What is the value of the word MATH?

Word	Value
TOTE	18
TEAM	168
MOM	49
HOME	70
MATH	?

Solution :

$$\begin{aligned} MOM &\equiv 49 \\ M \times O \times M &= 49 \\ M^2 &= \frac{49}{O} \\ M &= \sqrt{\frac{49}{O}} = \frac{7}{\sqrt{O}} \quad \Rightarrow \quad \begin{cases} M = 7, & \text{if } \sqrt{O} = 1 \ (O = 1) \\ M = 1, & \text{if } \sqrt{O} = 7 \ (O = 49) \end{cases} \end{aligned}$$

But $TOTE \equiv 18 < 49 \quad \Rightarrow \quad O$ *cannot be* 49*, thus* $O = 1$*, hence* $M = 7$.

$TOTE \equiv 18 \quad \Rightarrow \quad T^2 \times 1 \times E = 18 (O = 1) \quad \Rightarrow \quad T^2 = \frac{18}{E}$

If $E = 2$, $T^2 = 9 \ \Rightarrow \ T = \sqrt{9} = 3$

$T^2 = \frac{18}{E}$

- *If* $E = 2$; $T^2 = \frac{18}{2} \Rightarrow T = 3$
- *If* $T = 1$; $E = 18$, *but* T *can not be* 1 *since* $0 = 1$ *already;*

Thus $T = 3$, $E = 2$

$TEAM \equiv 168$

$$T \times E \times A \times M = 168$$
$$3 \times 2 \times A \times 7 = 168$$
$$A = \frac{168}{42}$$

$HOME \equiv 70$

$$H \times O \times M \times E = 70$$
$$H \times 1 \times 7 \times 2 = 70$$
$$H = \frac{70}{14} = 5$$

Thus,

$$\begin{aligned} MATH &\equiv M \times A \times T \times H \\ &= 7 \times 4 \times 3 \times 5 \\ &= 420 \end{aligned}$$

Therefore, $MATH \equiv 420$ ■

Qn. 13. In the magic square, all rows, columns and diagonals have the same sum. The magic square uses each of the numbers from -6 to $+2$ only. What is the value of $a + b + c + d + e$?

$+1$	a	d
-4	b	c
-3	e	-5

Solution : *The first column has sum*
column1: $+1 - 4 - 3 = -6$ *(sum of all columns or rows or diagonal)*

Diagonal: $+1 + b - 5 = -6 \Rightarrow b = -6 - 1 + 5 = -2$

row2: $-4 + b + c = -6 \Rightarrow -4 - 2 + c = -6 \Rightarrow c = 0$

row3: $-3 + e - 5 = -6 \Rightarrow e = 2$

column2: $a + b + e = -6 \Rightarrow a - 2 + 2 = -6 \Rightarrow a = -6$

column3: $d + c - 5 = -6 \Rightarrow d + 0 - 5 = -6 \Rightarrow d = -1$

$a = -6$, $b = -2$, $c = 0$, $d = -1$, $e = 2$
Thus $a + b + c + d + e = -6 - 2 + 0 - 1 + 2 = -9 + 2 = -7$.

■

Qn. 14. Three pineapples are weighted two at a time in all possible ways. The weights of the pairs of pineapples are 12 kg, 13 kg and 15 kg. What is the weight of the least heavy pineapple?

Solution : *Change "highest" to "least heavy" Let the weights be* A, B, C *in increasing order of weight.*

The possible pairs are; (A, B), (A, C), (B, C)

$$\begin{aligned} \textit{Least heavy}: \quad A+B &= 12 \quad (1.1)\\ \textit{Most heavy}: \quad B+C &= 15 \quad (1.2)\\ \textit{Thus} \quad A+C &= 13 \quad (1.3)\end{aligned}$$

$(1.3)-(1.1)$ *gives* $C-B=13-12=1 \Rightarrow C=B+1$.

Thus $B+c=15 \Rightarrow B+(B+1)=15$, *so* $2B=14 \Rightarrow B=7$

$C=B+1 \Rightarrow C=7+1=8$

$A+B=12 \Rightarrow A=12-B=12-7=5$

$A=5$, $B=7$, $C=8$. *Least heavy pineapple weighs* $5kg$.

■

Qn. 15. Chris and Farida want to buy the same book. Chris has $\frac{3}{4}$ of the cost of the book and Farida has half of the cost of the book. If the book was Shs. $30,000$ cheaper, then together they would have exactly enough money to buy 2 copies of the book. What is the original price of the book?

Solution : *Let the cost of the book be c, then Chris has* $\frac{3}{4}c$ *and Faridah has* $\frac{c}{2}$, *such that the total money is given by* $\frac{3}{4}c+\frac{c}{2}=\frac{5}{4}c$

If the book was $30,000$ *cheaper, then it would cost* $c-30,000$. *To buy two books, it would cost* $2(c-30,000)$. *Chris and Farida's money is enough for the two books, and so*

$$\frac{5}{4}c=2(c-30,000) \quad \Rightarrow \quad c=80,000$$

■